


# Hivemind

## Three control-plane attacks

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# Last week → today

| Last week's questions | Current status |
|---|---|
| Can we force leadership without knowing the group key K? | Ongoing |
| Crawl a live swarm every d seconds and record who has advertised. | No live averaging swarm found. Need a known group key to crawl. |
| Measure lookup time and the shortest advertised deadline. | Local tests using RIPE Atlas ping RTTs: Sweden, US, Brazil, India, Australia. |

# The delay profile: RIPE Atlas anchor RTTs

RIPE Atlas is a global measurement network whose anchor hosts continuously ping each other; we took the median of three ICMP replies per directed pair.

| from \ to | SE | US | BR | IN | AU |
|---|---|---|---|---|---|
| Stockholm | 9 | 99 | 209 | 147 | 298 |
| New York | 99 | 13 | 125 | 285 | 240 |
| São Paulo | 209 | 111 | 16 | 228 | 305 |
| Mumbai | 147 | 240 | 228 | 23 | 468 |
| Sydney | 299 | 239 | 305 | 503 | 12 |

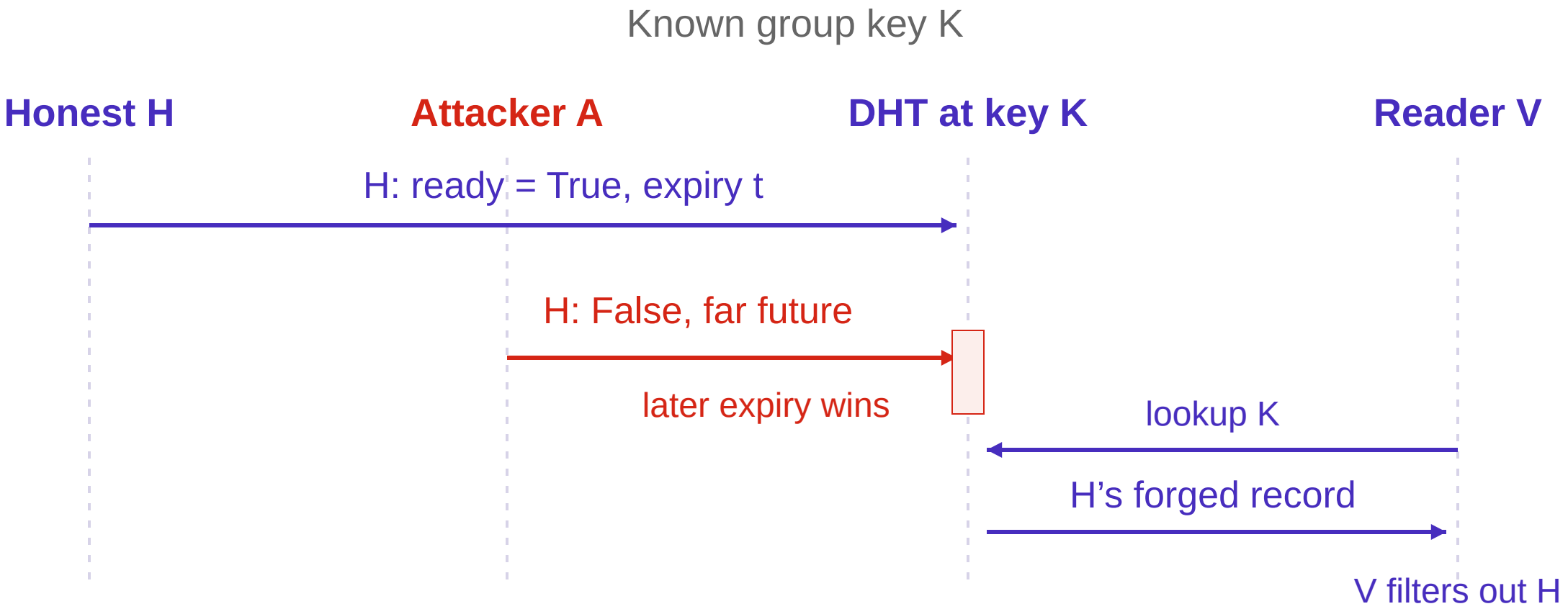
**233 ms**
median between regions

**99 – 503 ms**
range across the 20 links

Each region holds 5–6 of the 28 nodes in the swarm

Link RTTs, not DHT operation times · measured 30 Sep 2026 · profile saved as ripe_anchor_rtt_20260930.json

# 1. Forged withdrawal




H stays online, but disappears from group discovery.

H’s refresh expires sooner; the DHT keeps the record that expires later.

# Withdrawal: two separate tests

## 1 Can an attacker hide honest peers?

12 or 40 honest advertisers, plus one attacker.

Check whether forged withdrawals hide peers, even after they advertise again.

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## 2 Does hiding a peer make the target join the attacker?

Use RIPE Atlas delays in a local swarm.

Compare who the target joins, with and without a forged withdrawal.

# Withdrawal: results

| Check | Observed result |
|---|---|
| Targeted peers hidden | 39/39 (40 honest peers) |
| Refreshes that restored the peer | 0/30 attempts |
| Target joined attacker, with withdrawal | 0/10 trials |
| Target joined attacker, without withdrawal | 0/10 trials |

Group capture was tested locally with delays derived from RIPE Atlas pings.

Exact tables → A1

# 2. Deadline inflation

Keep H visible, but move its deadline behind the attacker's.

**Before**

| Peer | Advertised deadline |
|---|---|
| **H** | now + 5 s |
| **A** | now + 6 s |

V would contact H

forge H's expiration

**After forged write**

| Peer | Advertised deadline |
|---|---|
| **A** | now + 6 s |
| **H** | now + 1 hour |

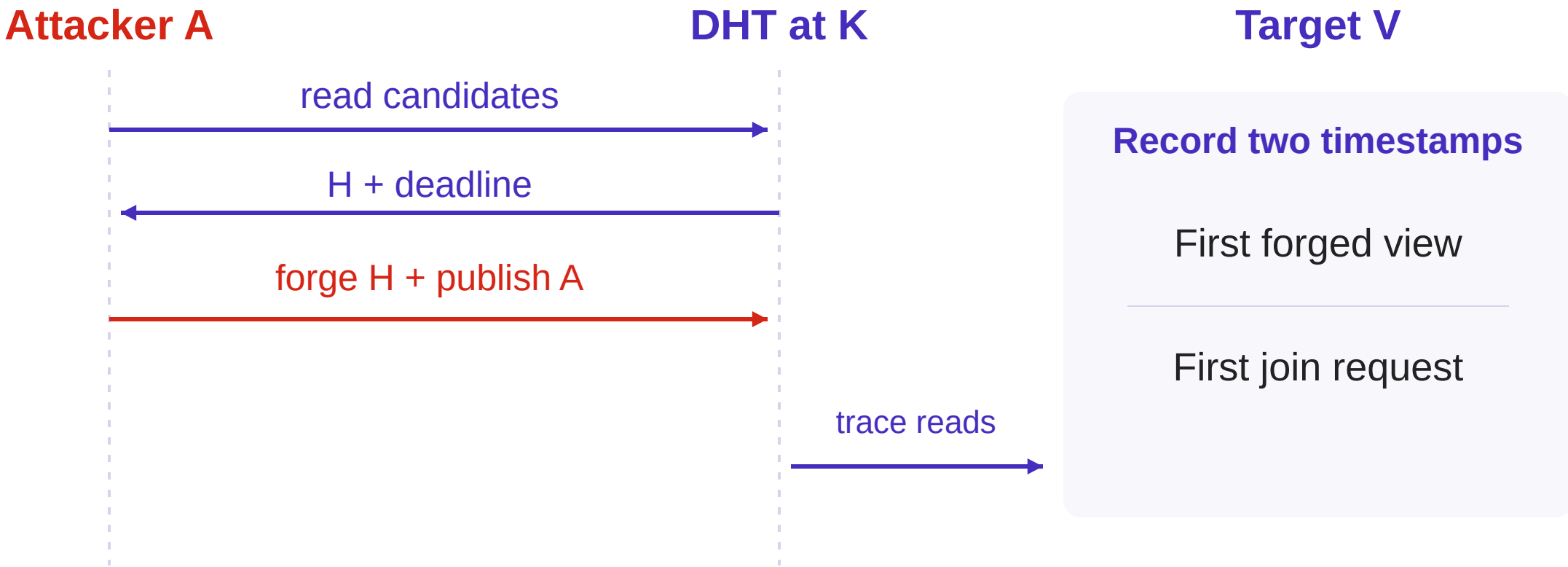
V may contact A

Only if V reads the changed record before choosing whom to join.

Illustrative times · H's ready flag stays True · no crafted DHT identity

# Inflation: measure the race to V’s view

A’s own deadline stays later than H’s; only H’s record is forged.




Compare timestamps: forged view before first join request?

28 local DHT nodes · 5 RTT origins · 5–7 s / 11–15 s windows · paired control

# Accepted writes did not change V’s join

| Measured event | Attack trials |
|---|---|
| DHT accepted the forged record | **10 / 10** |
| Forged view recorded before V’s first request | **0 / 10** |
| V joined the attacker | **0 / 10** |

No-forgery control: 0 / 10 targets joined A.

**In 7/10 trials:** V picked a peer → forged view appeared later

One reactive, single-peer policy; other strategies remain untested.

Exact tables → A2

# 3. Deadline undercutting

Read every advertised deadline, then declare one step earlier than all of them.

### Advertised at K

| Peer | Deadline |
|---|---|
| **H1** | now + 11.4 s |
| **H2** | now + 12.8 s |
| **H3** | now + 13.8 s |
| **A** | now + 11.4 s − 1 ulp |

undercut the earliest →

**Leader election**

earliest deadline wins

A leads; A admits whom it likes

Cost to A: one identity,
no compute, no crafted node ID

One ulp is $2.4\times10^{-7}$ s here · enough to win the comparison, no advertisement life lost

# Undercutting: two experiments

Same key, node IDs and honest schedule; only the attacker's region changes.

**Read ads at K** → **Undercut earliest** → **Publish own ad** → **Wait for joins**

### Experiment 1 · where the attacker sits

5 origins: Stockholm, New York, São Paulo, Mumbai, Sydney · RTTs from RIPE Atlas pings

2 honest windows: 5–7 s and 11–15 s

3 seeds each → 30 runs

### Experiment 2 · is the failure ours?

Our attacker skips any ad with under 3.5 s left (request_timeout 2.5 + our own 1.0 margin)

Set both that margin and the 0.3 s confirmation wait to zero · Stockholm, the cheapest origin

28 local DHT nodes, 5–6 per region · above bucket_size 20, so lookups must route

Injected delays on one machine · no loss, churn or connection setup · key known to A

# The window decides, not the attacker

| Attacker origin | 11–15 s window | 5–7 s window |
|---|---|---|
| Stockholm | **3 / 3** | **0 / 3** |
| New York | **3 / 3** | **0 / 3** |
| São Paulo | **3 / 3** | **0 / 3** |
| Mumbai | **3 / 3** | **0 / 3** |
| Sydney | **3 / 3** | **0 / 3** |

**Location changes cost, not outcome:** read + write 2.1 s from Stockholm, 3.7 s from Sydney

**Removing our own caution:** published 1/3 → 3/3, led 0/3 → 1/3, captured 0/3 either way

Figure: results/step19_region_sweep.png · raw runs: step19_wan_delay_*_r5_*.csv

# 4. One group, conflicting reducer indices

**Precondition: A is already the group leader.**




different
ordered rosters

Same group ID
Same four peers

Reducer index

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| A's own view | A | F1 | F2 | F3 |
| Sent to F1 | F1 | A | F2 | F3 |
| Sent to F2 | F2 | F1 | A | F3 |
| Sent to F3 | F3 | F1 | F2 | A |

Each recipient claims A's reducer slot (index 0 in this example).

Equal-sized slices can be mixed; followers do not cross-check one common roster.

# Roster test: change only the ordering

## 1 Same inputs

| | | | | |
|---|---|---|---|---|
| A | 0 | 1 | 2 | 4 |
| F1 | 0 | 1 | 2 | 4 |
| F2 | 0 | 1 | 2 | 4 |
| F3 | 0 | 1 | 2 | 4 |

Four real local averagers

## 2 Two conditions

**One common roster**

5 paired trials

**Different roster per peer**

## 3 Check outputs

**Expected average**

| | | | |
|---|---|---|---|
| 0 | 1 | 2 | 4 |

Did it complete?

Is the tensor correct?

Were failures recorded?

Real DHT + join streams + stock all-reduce · leadership arranged in advance

# Wrong tensors, successful steps

## One recorded trial

Correct average

| 0 | 1 | 2 | 4 |
|---|---|---|---|

One attacked peer’s output

| 1.75 | 0.75 | 1.5 | 3.0 |
|---|---|---|---|

## Across five paired trials

Corrupted trials

| | |
|---|---|
| Control | **0 / 5** |
| Attack | **5 / 5** |

**20/20 attacked outputs wrong**

All 40 steps completed; tested failure indicators stayed clear.

Tensor corruption measured locally; downstream training damage was not tested.

Exact tables → A3

# A1. Withdrawal discovery tests

One ordinary random-ID writer · counts below matched in both repeats

| Honest peers | Targets / run | Hidden / run | Others visible | Repeats |
|---|---|---|---|---|
| 12 | 1 | 1 | 11 / 11 | 2 |
| 12 | 3 | 3 | 9 / 9 | 2 |
| 12 | 6 | 6 | 6 / 6 | 2 |
| 12 | 11 | 11 | 1 / 1 | 2 |
| 40 | 1 | 1 | 39 / 39 | 2 |
| 40 | 20 | 20 | 20 / 20 | 2 |
| 40 | 39 | 39 | 1 / 1 | 2 |

Separate refresh test: 0/30 attempts restored H in each swarm size.

Source: step8_withdrawal_phase2.csv and step8_withdrawal_phase2_n40.csv

Exact tables → A2

# A2. Forged-record trials

Targets that joined A / trials · one seed per region and deadline window

| RTT origin | Window | Control | Withdrawal | Inflation |
|---|---|---|---|---|
| Sweden | 5-7 s | 0 / 1 | 0 / 1 | 0 / 1 |
| United States | 5-7 s | 0 / 1 | 0 / 1 | 0 / 1 |
| Brazil | 5-7 s | 0 / 1 | 0 / 1 | 0 / 1 |
| India | 5-7 s | 0 / 1 | 0 / 1 | 0 / 1 |
| Australia | 5-7 s | 0 / 1 | 0 / 1 | 0 / 1 |
| Sweden | 11-15 s | 0 / 1 | 0 / 1 | 0 / 1 |
| United States | 11-15 s | 0 / 1 | 0 / 1 | 0 / 1 |
| Brazil | 11-15 s | 0 / 1 | 0 / 1 | 0 / 1 |
| India | 11-15 s | 0 / 1 | 0 / 1 | 0 / 1 |
| Australia | 11-15 s | 0 / 1 | 0 / 1 | 0 / 1 |

Forgery accepted in 20/20 attack trials (29/29 writes); all 90 honest steps completed.

28-node local RTT emulation · source: step20 sweep_20261002_a analysis + event CSVs

# A3. Reducer-index trials

Four peers · five paired local trials · identical input [0, 1, 2, 4]

| Trial | Control correct | Attack correct | Max. abs. error | Steps |
|---|---|---|---|---|
| 1 | 4 / 4 | 0 / 4 | 3.00 | 8 / 8 |
| 2 | 4 / 4 | 0 / 4 | 3.00 | 8 / 8 |
| 3 | 4 / 4 | 0 / 4 | 3.00 | 8 / 8 |
| 4 | 4 / 4 | 0 / 4 | 2.25 | 8 / 8 |
| 5 | 4 / 4 | 0 / 4 | 3.00 | 8 / 8 |

No bans, fallback elements, RPC/all-reduce errors, timeouts, or hard cancellations.

Source: step17_roster_live_rounds_confirm_20260928_confirm_t5_s1717.csv